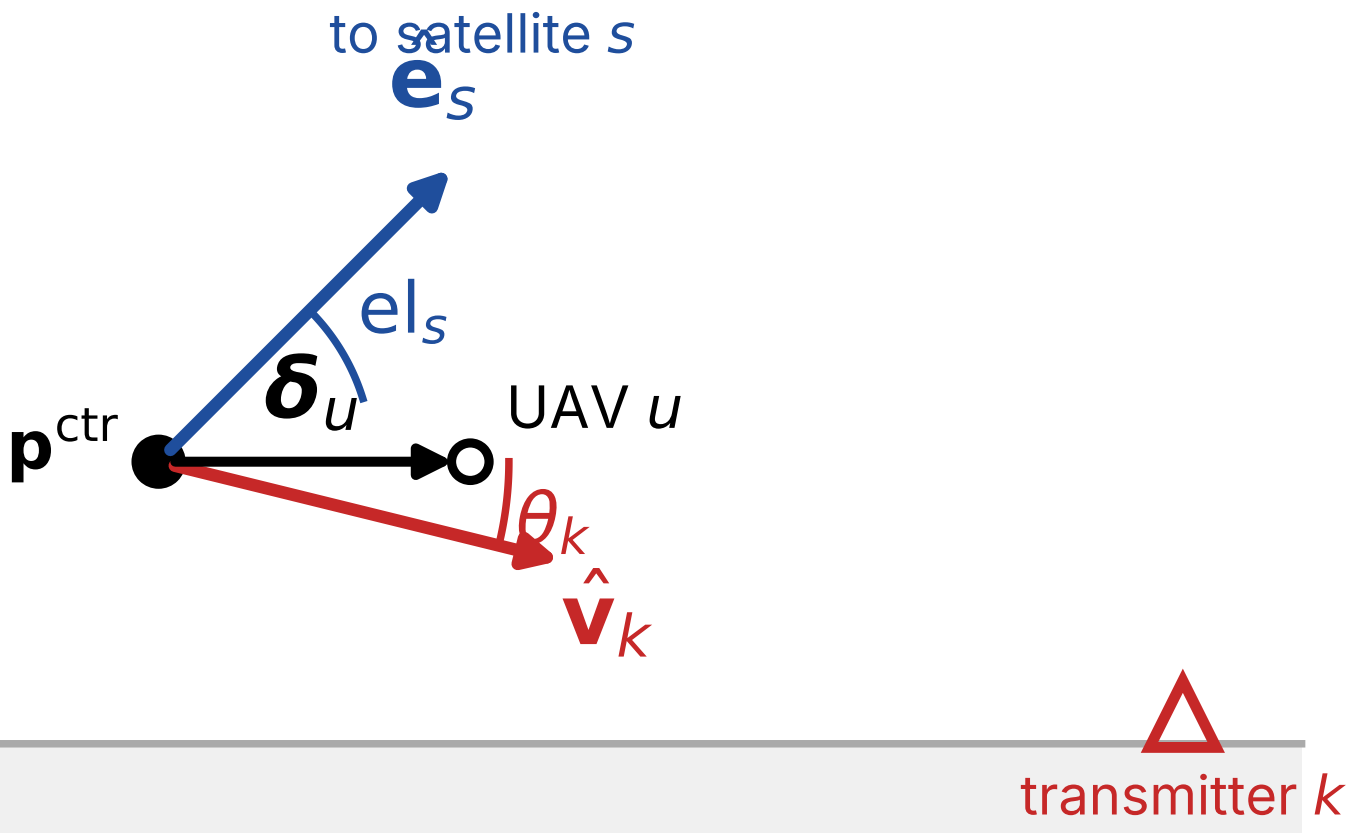

$$\varepsilon_{u,s} = (\hat{\mathbf{e}}_s - \hat{\mathbf{v}}_k) \cdot \boldsymbol{\delta}_u$$
$$\approx -(\hat{\mathbf{a}}_s \cdot \boldsymbol{\delta}_u)(\cos\theta_k - \cos\mathrm{el}_s) < 0$$